A Conditional Positive-Loss Certificate for Half-Relaxed Parallel Tensor ALS

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1 Theoretical Setup

1.1 Tensor model and objective

Fix constants $\kappa \geq 1$ and $q > 0$, and let r, n, k be positive integers. For each mode, let $\bar{A} = [\bar{a}_1, \dots, \bar{a}_r] \in \mathbb{R}^{n \times r}$, with analogous matrices $\bar{B}$ and $\bar{C}$. For $\bar{M} \in \{\bar{A}, \bar{B}, \bar{C}\}$, write

$$D_{\bar{M}} = \operatorname{diag}(\|\bar{m}_1\|_2, \dots, \|\bar{m}_r\|_2), \qquad \widetilde{M} = \bar{M} D_{\bar{M}}^{-1}.$$

Set $\rho = r^{-q}$. Independently over $j \in [r]$ and the three modes, draw

$$\xi_j^a, \xi_j^b, \xi_j^c \sim \mathcal{N}\left(0, \frac{\rho^2}{n} I_n\right),$$

and define

$$a_j = \bar{a}_j + \xi_j^a, \qquad b_j = \bar{b}_j + \xi_j^b, \qquad c_j = \bar{c}_j + \xi_j^c.$$

The realized target tensor is

$$T = \sum_{j=1}^{r} a_j \otimes b_j \otimes c_j \in \mathbb{R}^{n \times n \times n}.$$

For factor matrices $X = [x_1, \dots, x_k]$, $Y = [y_1, \dots, y_k]$, and $Z = [z_1, \dots, z_k]$, define

$$\widehat{T}(X, Y, Z) = \sum_{i=1}^{k} x_i \otimes y_i \otimes z_i, \qquad \mathcal{L}(X, Y, Z) = \|T - \widehat{T}(X, Y, Z)\|_F^2.$$

At time t, abbreviate $\widehat{T}_t = \widehat{T}(X_t, Y_t, Z_t)$. Fix one componentwise Khatri–Rao ordering, put

$$U_t^x = Z_t \odot Y_t, \qquad U_t^y = Z_t \odot X_t, \qquad U_t^z = Y_t \odot X_t,$$

and let $T_{(1)}, T_{(2)}, T_{(3)}$ be the corresponding matricizations.

1.2 Half-relaxed parallel ALS and product gauge

All three least-squares candidates use the old iterate and are computed in parallel:

$$\begin{aligned} X_{t+1}^{\mathrm{ls}} &= T_{(1)} U_t^x \left((U_t^x)^\top U_t^x \right)^\dagger, \\ Y_{t+1}^{\mathrm{ls}} &= T_{(2)} U_t^y \left((U_t^y)^\top U_t^y \right)^\dagger, \\ Z_{t+1}^{\mathrm{ls}} &= T_{(3)} U_t^z \left((U_t^z)^\top U_t^z \right)^\dagger. \end{aligned}$$

The Moore–Penrose pseudoinverse selects the minimum-Frobenius-norm least-squares candidate. With relaxation parameter $\eta = 1/2$, form

$$X_{t+1}^{\mathrm{raw}} = \tfrac{1}{2}(X_t + X_{t+1}^{\mathrm{ls}}), \quad Y_{t+1}^{\mathrm{raw}} = \tfrac{1}{2}(Y_t + Y_{t+1}^{\mathrm{ls}}), \quad Z_{t+1}^{\mathrm{raw}} = \tfrac{1}{2}(Z_t + Z_{t+1}^{\mathrm{ls}}).$$

There is no fixed-subspace constraint, clipping, regularizer, adaptive preconditioner, restart, or early stopping. Sequential Gauss–Seidel ALS and unrelaxed simultaneous ALS are not covered.

After each relaxed sweep, apply the following deterministic gauge to every component. If $u = \|x_i\|_2$, $v = \|y_i\|_2$, and $w = \|z_i\|_2$ are positive, let $g = (uvw)^{1/3}$ and replace

$$(x_i, y_i, z_i) \quad \text{by} \quad (gx_i/u, gy_i/v, gz_i/w).$$

The three scale factors have product one. If $uvw = 0$, replace the component triple by $(0, 0, 0)$. The resulting factors are denoted $(X_{t+1}, Y_{t+1}, Z_{t+1})$. The initial state is the raw Gaussian draw specified below; the gauge is first applied after the first sweep.

1.3 Primitive assumptions

Assumption 1 (Ambient dimension)**.** *The integers r, n are positive and*

$$n \geq C_{\mathrm{dim}}(\kappa, q) r^4 \log r,$$

where $C_{\mathrm{dim}}(\kappa, q) > 0$ is independent of r, n, k and of the deterministic base triple.

Assumption 2 (Full superlinear rank window)**.** *The integer k satisfies*

$$r < k \leq r^{5/4}.$$

Thus the universal overparameterization exponent is $1/4$.

Assumption 3 (Well-conditioned deterministic bases)**.** *For every $\bar{M} \in \{\bar{A}, \bar{B}, \bar{C}\}$ and every $j \in [r]$,*

$$\kappa^{-1} \leq \|\bar{m}_j\|_2 \leq \kappa,$$

and all r singular values of the rectangular matrix $\widetilde{M} \in \mathbb{R}^{n \times r}$—equivalently, the square roots of all r eigenvalues of $\widetilde{M}^{\mathsf{T}}\widetilde{M}$—lie in $[\kappa^{-1}, \kappa]$. This is the full-column-rank rectangular singular-value convention used throughout.

Assumption 4 (Independent Gaussian smoothing)**.** *The exponent $q > 0$ is fixed, $\rho = r^{-q}$, and the $3r$ perturbation vectors have the mutually independent Gaussian law specified above.*

Assumption 5 (Independent Gaussian initialization)**.** *All $3nk$ entries of X_0, Y_0, Z_0 are mutually independent $\mathcal{N}(0, 1/n)$ variables and are independent of all smoothing variables.*

All probabilities below refer to this joint smoothing-and-initialization law, conditional on the deterministic base triple.

1.4 Formalized goal

The goal is an explicitly conditional theorem. We seek constants depending only on (κ, q) for which membership in the four-clause certificate defined in Section 2 implies that the objective has a finite limit and that this limit is a positive constant fraction of $\|T\|_F^2$. The target fraction is

$$\epsilon = \left(\frac{\delta - L_P - \zeta}{\kappa^6 C_T} \right)^2.$$

The theorem does not assert that the certificate is nonempty, and it does not assert any positive lower bound on the probability of the certificate. Establishing such a uniform probability bound is the residual source-level gap.

2 Preliminaries

Assumption 3 makes each deterministic base matrix full column rank. Define the fixed left coordinate maps

$$\Lambda_A = (\bar{A}^\top \bar{A})^{-1}\bar{A}^\top, \qquad \Lambda_B = (\bar{B}^\top \bar{B})^{-1}\bar{B}^\top, \qquad \Lambda_C = (\bar{C}^\top \bar{C})^{-1}\bar{C}^\top,$$

and their modewise tensor product

$$Q = \Lambda_A \otimes \Lambda_B \otimes \Lambda_C.$$

The map Q is used in the theorem to compare the ambient residual with a coefficient-space deficit. Both coefficient and ambient tensor spaces carry their Frobenius norms.

For every component and time, set

$$\alpha_i^t = \Lambda_A x_{i,t}, \qquad \beta_i^t = \Lambda_B y_{i,t}, \qquad \gamma_i^t = \Lambda_C z_{i,t}, \qquad p_{i,t} = \alpha_i^t \otimes \beta_i^t \otimes \gamma_i^t.$$

With $e_1, \ldots, e_r$ the standard basis of $\mathbb{R}^r$, define

$$D_r = \sum_{j=1}^{r} e_j \otimes e_j \otimes e_j, \qquad \|D_r\|_F = \sqrt{r},$$

and

$$C_t = \sum_{i=1}^{k} p_{i,t}, \qquad \mathcal{S}_t = \operatorname{span}\{p_{i,t} : i \in [k]\}, \qquad P_t = \operatorname{Proj}_{\mathcal{S}_t}.$$

Here P_t is the orthogonal projector in coefficient Frobenius geometry, including $P_t = 0$ if $\mathcal{S}_t = \{0\}$. Finally, put

$$\Delta_0 = \operatorname{dist}_F(D_r, \mathcal{S}_0), \qquad E_\rho = QT - D_r.$$

These quantities are all defined from the setting objects. In particular, the adaptive spaces $\mathcal{S}_t$ are outputs of the unconstrained ALS trajectory and are not required to remain in $\mathcal{S}_0$ or in any other fixed space.

For positive constants δ, L_P, ζ, C_T satisfying

$$L_P < \delta/4, \qquad \zeta < \delta/4,$$

let $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$ denote exactly the event on which the following four clauses hold:

1. *Normalized entry deficit:*
$$\Delta_0 \geq \delta \|D_r\|_F = \delta\sqrt{r}.$$
2. *Finite adaptive-projector path:*
$$\sum_{t=0}^{\infty} \|P_{t+1} - P_t\|_{\mathrm{op}} \leq L_P.$$
3. *Unsquared finite represented-tensor variation:*
$$\sum_{t=0}^{\infty} \|\widehat{T}_{t+1} - \widehat{T}_t\|_F < \infty.$$

4. *Relative smoothing and target scale:*

$$\|E_\rho\|_F \le \zeta\|D_r\|_F, \qquad \|T\|_F \le C_T\|D_r\|_F,$$

where $C_T = C_T(\kappa, q)$ is independent of r, n, k and of the admissible deterministic base triple.

These four clauses are conditional hypotheses of the main theorem, not primitive assumptions and not claimed consequences of Assumptions 1–5. They include no Gram condition, factor bound, fixed-subspace restriction, basin, trapping event, persistent residual premise, positive-loss premise, or assertion about their own probability.

3 Main Theorem

Theorem 3.1 (Conditional positive limiting loss)**.** *Fix $\kappa \ge 1$ and $q > 0$. Let $r_0, C_{\dim}, \delta, L_P, \zeta, C_T$ be positive quantities that may depend only on (κ, q), with $r_0 \in \mathbb{N}$ and*

$$L_P < \delta/4, \qquad \zeta < \delta/4.$$

Define

$$\epsilon = \left(\frac{\delta - L_P - \zeta}{\kappa^6 C_T}\right)^2 > 0.$$

For every $r \ge r_0$, every n, k satisfying Assumptions 1 and 2, every deterministic base triple satisfying Assumption 3, and the half-relaxed parallel ALS trajectory under Assumptions 4 and 5, one has the event inclusion

$$\mathsf{C}_2(\delta, L_P, \zeta, C_T) \subseteq \left\{ \begin{array}{l} \lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \text{ exists and is finite,} \\ \lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \ge \epsilon\|T\|_F^2 \end{array} \right\}.$$

This is a deterministic implication on each realization in the displayed event. It makes no assertion that C_2 is nonempty and no assertion that it has positive probability.

There are no hidden constants in the displayed loss fraction. Its exposed dependence is exactly on $\kappa, \delta, L_P, \zeta, C_T$; r, n, k enter only through the theorem scope and certificate objects. The probability mode is event inclusion under the joint law, the residual floor used in the proof is all-time, the conclusion is asymptotic, and all residuals and objectives use the ambient Frobenius norm.

Separately from the preceding positive-Gaussian-smoothing event inclusion, consider the deterministic algebraic zero-smoothing baseline in a tall ambient space $n \ge r$: each of $\bar{A}, \bar{B}, \bar{C}$ has orthonormal columns, every perturbation vector is set to zero, and hence $T = \sum_{j=1}^r \bar{a}_j \otimes \bar{b}_j \otimes \bar{c}_j$. In this baseline, the ambient-to-coefficient map is generally rectangular, but it satisfies the well-typed relations

$$QT = D_r, \qquad E_\rho = 0, \qquad \|Q\|_{\mathrm{op}} = 1, \qquad \|T\|_F = \|D_r\|_F.$$

For any half-relaxed parallel ALS trajectory for this deterministic target that satisfies the normalized entry-deficit, finite projector-path, and finite represented-tensor-variation clauses, the same argument gives the stronger baseline conclusion

$$\lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \ge (\delta - L_P)^2\|T\|_F^2.$$

This last statement is an algebraic noiseless comparison, not an event under Assumption 4 or the positive-smoothing joint law used in the first part of the theorem. In particular, because the conditional implication holds for every positive choice with the stated margins, it implies the existential constant statement in the formalized goal, with all theorem constants depending only on (κ, q).

The unresolved source-level problem is to prove a uniform bound $\mathbb{P}[\mathsf{C}_2(\delta, L_P, \zeta, C_T)] \ge p_0(\kappa, q) > 0$ for some admissible constants. Theorem 3.1 neither proves nor assumes such a bound.

4 Proof Sketch

The argument has five parts. First, base conditioning gives $\|\Lambda_M\|_{\mathrm{op}} \le \kappa^2$ in each mode and therefore $\|Q\|_{\mathrm{op}} \le \kappa^6$. Tensor-product linearity and the product-one gauge then yield the exact coefficient identity

$$Q(T - \widehat{T}_t) = D_r + E_\rho - C_t, \qquad C_t \in \mathcal{S}_t.$$

Second, orthogonal-projector geometry gives the one-step comparison

$$\mathrm{dist}_F(D_r, \mathcal{S}_{t+1}) \ge \mathrm{dist}_F(D_r, \mathcal{S}_t) - \|P_{t+1} - P_t\|_{\mathrm{op}} \|D_r\|_F.$$

The first two certificate clauses telescope this inequality over every finite horizon and leave the all-time reserve $(\delta - L_P)\|D_r\|_F$.

Third, distance to a fixed subspace is 1-Lipschitz. The smoothing residual therefore costs at most $\zeta\|D_r\|_F$. The exact coordinate identity, $\|Q\|_{\mathrm{op}} \le \kappa^6$, and $\|T\|_F \le C_T\|D_r\|_F$ give, for every t,

$$\|T - \widehat{T}_t\|_F \ge \frac{\delta - L_P - \zeta}{\kappa^6 C_T} \|T\|_F.$$

Fourth, the third certificate clause makes the represented tensors Cauchy: the norm of every tail difference is bounded by the corresponding tail of the absolutely summable increment series. Hence $\widehat{T}_t$ converges, and continuity of squared Frobenius distance gives a finite objective limit.

Finally, the residual floor is squared only after checking its nonnegative margin, and the resulting all-time scalar lower bound passes to the existing finite limit. This produces the event inclusion without conditioning on the event's probability.

For the separate deterministic zero-smoothing baseline, take tall column-orthonormal bases and set every perturbation vector to zero. Then $QT = D_r$, $E_\rho = 0$, $\|Q\|_{\mathrm{op}} = 1$, and $\|T\|_F = \|D_r\|_F$. Thus the same-target identity and the operator comparison give

$$\|T - \widehat{T}_t\|_F \ge \|Q(T - \widehat{T}_t)\|_F = \|D_r - C_t\|_F \ge (\delta - L_P)\|T\|_F.$$

Squaring and passing to the finite-variation limit yields the stronger baseline factor $(\delta - L_P)^2$. This algebraic baseline lies outside the positive-Gaussian-smoothing probability statement above.

A Proof Details

A.1 Coordinate maps, product gauge, and the same-target identity

Lemma A.1 (Conditioned left coordinate maps). *Under Assumption 3, every $\bar{M} \in \{\bar{A}, \bar{B}, \bar{C}\}$ has full column rank, and its coordinate map*

$$\Lambda_M = (\bar{M}^\top \bar{M})^{-1} \bar{M}^\top$$

satisfies

$$\sigma_{\min}(\bar{M}) \geq \kappa^{-2}, \qquad \Lambda_M \bar{M} = I_r, \qquad \|\Lambda_M\|_{\mathrm{op}} = \sigma_{\min}(\bar{M})^{-1} \leq \kappa^2.$$

Proof. The factorization $\bar{M} = \widetilde{M} D_{\bar{M}}$ and Assumption 3 give

$$\sigma_{\min}(\widetilde{M}) \geq \kappa^{-1}, \qquad \sigma_{\min}(D_{\bar{M}}) = \min_{j \in [r]} \|\bar{m}_j\|_2 \geq \kappa^{-1}.$$

Thus, for every $v \in \mathbb{R}^r$,

$$\begin{aligned} \|\bar{M} v\|_2 &= \|\widetilde{M} D_{\bar{M}} v\|_2 \\ &\geq \sigma_{\min}(\widetilde{M}) \|D_{\bar{M}} v\|_2 \\ &\geq \kappa^{-2} \|v\|_2. \end{aligned}$$

Consequently $\sigma_{\min}(\bar{M}) \geq \kappa^{-2} > 0$, so $\bar{M}$ has full column rank and $\bar{M}^\top \bar{M}$ is invertible. Direct multiplication then gives $\Lambda_M \bar{M} = I_r$.

For completeness, if $\bar{M} = U \Sigma V^\top$ is an economy singular-value decomposition, then $U^\top U = V^\top V = I_r$ and Σ is positive diagonal. Hence

$$(\bar{M}^\top \bar{M})^{-1} \bar{M}^\top = V \Sigma^{-1} U^\top,$$

so its induced Euclidean operator norm is $\sigma_{\min}(\bar{M})^{-1}$. The preceding lower bound therefore yields $\|\Lambda_M\|_{\mathrm{op}} \leq \kappa^2$. This calculation remains valid at the non-strict conditioning boundary $\sigma_{\min}(\widetilde{M}) = \min_j \|\bar{m}_j\|_2 = \kappa^{-1}$. □

Lemma A.2 (Modewise tensor operator in Frobenius geometry). *Suppose Assumption 3 holds.*

For the setting-ordered map

$$Q = \Lambda_A \otimes \Lambda_B \otimes \Lambda_C$$

Lemma A.1 implies

$$Q(x \otimes y \otimes z) = (\Lambda_A x) \otimes (\Lambda_B y) \otimes (\Lambda_C z)$$

for every $x, y, z \in \mathbb{R}^n$, and

$$0 < \|Q\|_{\mathrm{op}} = \|\Lambda_A\|_{\mathrm{op}} \|\Lambda_B\|_{\mathrm{op}} \|\Lambda_C\|_{\mathrm{op}} \leq \kappa^6.$$

Consequently every $R \in \mathbb{R}^{n \times n \times n}$ obeys

$$\|QR\|_F \leq \kappa^6 \|R\|_F.$$

Proof. The pure-tensor formula is the defining action of the tensor product in the fixed mode order. To compute its norm, take full singular-value decompositions

$$\Lambda_M = U_M \Sigma_M V_M^\top, \qquad M \in \{A, B, C\},$$

where the outer matrices are square orthogonal and each Σ_M is rectangular diagonal. Under the canonical identification of a third-order array with the Hilbert tensor product of its mode spaces,

$$Q = (U_A \otimes U_B \otimes U_C)(\Sigma_A \otimes \Sigma_B \otimes \Sigma_C)(V_A^\top \otimes V_B^\top \otimes V_C^\top).$$

The outer maps are Frobenius isometries. Indeed, on pure tensors this follows from

$$\|u \otimes v \otimes w\|_F = \|u\|_2 \|v\|_2 \|w\|_2,$$

and expansion in orthonormal product bases extends the identity to the whole finite-dimensional tensor space. The middle map is diagonal in product singular-vector bases, with diagonal entries

$$\sigma_a(\Lambda_A)\sigma_b(\Lambda_B)\sigma_c(\Lambda_C).$$

Its largest singular value is therefore

$$\max_{a,b,c} \sigma_a(\Lambda_A)\sigma_b(\Lambda_B)\sigma_c(\Lambda_C) = \prod_{M \in \{A,B,C\}} \|\Lambda_M\|_{\mathrm{op}}.$$

Every Λ_M has rank $r \geq 1$ by Lemma A.1, so this product is positive. The same lemma bounds each factor by κ^2, giving

$$0 < \|Q\|_{\mathrm{op}} \leq (\kappa^2)^3 = \kappa^6.$$

The inequality for an arbitrary R is the definition of the induced operator norm. Notice that Q need not be injective on the ambient tensor space: a component of R may lie in $\ker Q$. The upper operator inequality remains valid and is exactly the direction used later. □

Lemma A.3 (Gauge invariance of ambient and coefficient components)**.** *Under Assumption 3 and Lemma A.2, let $(x, y, z) \in (\mathbb{R}^n)^3$, put*

$$p = (\Lambda_A x) \otimes (\Lambda_B y) \otimes (\Lambda_C z),$$

and let (x^+, y^+, z^+) be the result of the setting-defined componentwise gauge. Then

$$x^+ \otimes y^+ \otimes z^+ = x \otimes y \otimes z$$

and

$$(\Lambda_A x^+) \otimes (\Lambda_B y^+) \otimes (\Lambda_C z^+) = p.$$

Both statements hold in the positive-norm branch and in every zero-factor branch.

Proof. Let $u = \|x\|_2$, $v = \|y\|_2$, and $w = \|z\|_2$. If all three are positive, set $g = (uvw)^{1/3}$ and

$$s_x = g/u, \qquad s_y = g/v, \qquad s_z = g/w.$$

The gauge gives $(x^+, y^+, z^+) = (s_x x, s_y y, s_z z)$, and

$$s_x s_y s_z = \frac{g^3}{uvw} = 1.$$

Multilinearity yields

$$x^+ \otimes y^+ \otimes z^+ = (s_x s_y s_z)(x \otimes y \otimes z) = x \otimes y \otimes z.$$

Linearity of the three coordinate maps gives the same calculation in coefficient space:

$$(\Lambda_A x^+) \otimes (\Lambda_B y^+) \otimes (\Lambda_C z^+) = (s_x s_y s_z)p = p.$$

If $uvw = 0$, at least one of x, y, z is the zero vector. The original ambient rank-one tensor is therefore zero, and the corresponding coordinate vector is zero by linearity, so $p = 0$. The gauge replaces the triple by $(0, 0, 0)$, and both post-gauge tensors are also zero. The two cases exhaust all possible norm triples. Arbitrarily small positive norms do not create a new case: individual scale factors may be large, but their product remains exactly one. □

Proposition A.1 (Exact coordinate and same-target interface)**.** *Under Assumption 3 and Lemmas A.1–A.3, the map Q satisfies $0 < \|Q\|_{\mathrm{op}} \le \kappa^6$, the product gauge preserves every ambient and coefficient component tensor, and every integer $t \ge 0$ satisfies*

$$Q\widehat{T}_t = C_t \in \mathcal{S}_t, \qquad Q(T - \widehat{T}_t) = D_r + E_\rho - C_t.$$

These conclusions include the raw initialization, zero components, the zero coefficient span, linearly dependent features, and changes in $\dim(\mathcal{S}_t)$.

Proof. The operator bound and gauge assertions are Lemmas A.2 and A.3. For each $i \in [k]$, Lemma A.2 gives

$$Q(x_{i,t} \otimes y_{i,t} \otimes z_{i,t}) = (\Lambda_A x_{i,t}) \otimes (\Lambda_B y_{i,t}) \otimes (\Lambda_C z_{i,t}) = p_{i,t}.$$

Linearity over the finite CP sum therefore gives

$$Q\widehat{T}_t = \sum_{i=1}^{k} Q(x_{i,t} \otimes y_{i,t} \otimes z_{i,t}) = \sum_{i=1}^{k} p_{i,t} = C_t.$$

By definition, this finite sum belongs to $\mathcal{S}_t = \operatorname{span}\{p_{i,t} : i \in [k]\}$. The conclusion remains true if some features vanish or are linearly dependent. If they all vanish, then $C_t = 0 \in \mathcal{S}_t = \{0\}$.

The definition $E_\rho = QT - D_r$ is equivalent to $QT = D_r + E_\rho$. Subtracting the identity for $Q\widehat{T}_t$ gives

$$Q(T - \widehat{T}_t) = QT - Q\widehat{T}_t = D_r + E_\rho - C_t.$$

This is the coefficient image of the same realized ambient residual; no projected, whitened, base-only, or fixed-subspace surrogate has replaced it. At $t = 0$ the calculation applies directly to the ungauged Gaussian factors. At later times it applies to the stored factors, while Lemma A.3 shows that replacing a raw relaxed component by its gauged representative changes neither tensor sum. The calculation is pointwise in t and hence requires neither continuity nor constant rank of $\mathcal{S}_t$.

Thus the three modewise κ^2 bounds combine with no hidden constant,

$$\|Q\|_{\mathrm{op}} = \prod_{M \in \{A,B,C\}} \|\Lambda_M\|_{\mathrm{op}} \le \kappa^6,$$

and the exact same-target identity holds at every time.

For the separate deterministic zero-smoothing baseline, let $n \ge r$, let $\bar{A}, \bar{B}, \bar{C}$ have orthonormal columns, set every perturbation vector to zero, and hence take

$$T = \sum_{j=1}^{r} \bar{a}_j \otimes \bar{b}_j \otimes \bar{c}_j.$$

Then $\Lambda_M = \bar{M}^{\mathsf{T}}$ and $\Lambda_M \bar{m}_j = e_j$. Consequently

$$QT = D_r, \qquad E_\rho = 0, \qquad \|Q\|_{\mathrm{op}} = 1.$$

The r ambient rank-one summands are orthonormal, so $\|T\|_F = \sqrt{r} = \|D_r\|_F$. The exact identity therefore specializes to the well-typed coefficient-space relation

$$Q(T - \widehat{T}_t) = D_r - C_t, \qquad C_t \in \mathcal{S}_t.$$

When $n > r$, Q remains an ambient-to-coefficient partial isometry rather than an identity map. This deterministic algebraic baseline is separate from the positive-Gaussian-smoothing law in the main event-inclusion statement. All statements in this subsection are deterministic and pointwise. No certificate clause or probability conversion is used. □

A.2 Transport of the coefficient deficit along the projector path

Lemma A.4 (One-step transport under projector motion)**.** *For the coefficient Frobenius space and the setting-defined orthogonal projectors $P_t = \mathrm{Proj}_{\mathcal{S}_t}$, every integer $t \geq 0$ satisfies*

$$\mathrm{dist}_F(D_r, \mathcal{S}_{t+1}) \geq \mathrm{dist}_F(D_r, \mathcal{S}_t) - \|P_{t+1} - P_t\|_{\mathrm{op}} \|D_r\|_F.$$

No certificate clause is required for this one-step statement, and it remains valid for zero, full, stationary, or rank-changing subspaces.

Proof. Let $\mathcal{S}$ be a coefficient subspace and P its orthogonal projector. For every $y \in \mathcal{S}$, the vectors $(I - P)D_r \in \mathcal{S}^\perp$ and $PD_r - y \in \mathcal{S}$ are orthogonal. Pythagoras therefore gives

$$\|D_r - y\|_F^2 = \|(I - P)D_r\|_F^2 + \|PD_r - y\|_F^2 \geq \|(I - P)D_r\|_F^2.$$

Equality is attained at $y = PD_r$, so

$$\mathrm{dist}_F(D_r, \mathcal{S}) = \|(I - P)D_r\|_F.$$

Apply this identity at times t and $t + 1$. The exact relation

$$(I - P_t)D_r = (I - P_{t+1})D_r + (P_{t+1} - P_t)D_r$$

and the triangle inequality imply

$$\begin{aligned} \mathrm{dist}_F(D_r, \mathcal{S}_t) &= \|(I - P_t)D_r\|_F \\ &\leq \|(I - P_{t+1})D_r\|_F + \|(P_{t+1} - P_t)D_r\|_F \\ &\leq \mathrm{dist}_F(D_r, \mathcal{S}_{t+1}) + \|P_{t+1} - P_t\|_{\mathrm{op}} \|D_r\|_F. \end{aligned}$$

Rearrangement gives the claimed recurrence. If $\mathcal{S}_j = \{0\}$, then $P_j = 0$ and the distance is $\|D_r\|_F$; if $\mathcal{S}_j$ is the full coefficient space, then $P_j = I$ and the distance is zero. A stationary projector contributes zero charge, while any rank change is charged by its actual projector difference. No rank-continuity premise is used. □

Proposition A.2 (Finite-path preservation of the coefficient deficit)**.** *Suppose clauses 1 and 2 of $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$ hold, with $\delta > 0$, $L_P > 0$, and $L_P < \delta/4$. Then, for every integer $t \geq 0$,*

$$\mathrm{dist}_F(D_r, \mathcal{S}_t) \geq \Delta_0 - \|D_r\|_F \sum_{s=0}^{t-1} \|P_{s+1} - P_s\|_{\mathrm{op}} \geq (\delta - L_P)\|D_r\|_F.$$

For $t = 0$ the sum is empty, and

$$\delta - L_P > \frac{3\delta}{4} > 0.$$

Proof. We first prove the finite-horizon inequality by induction. At $t = 0$, the empty-sum convention gives

$$\mathrm{dist}_F(D_r, \mathcal{S}_0) = \Delta_0,$$

so equality holds. If the claim holds at time t, then Lemma A.4 gives

$$\begin{aligned}\mathrm{dist}_F(D_r, \mathcal{S}_{t+1}) &\geq \mathrm{dist}_F(D_r, \mathcal{S}_t) - \|P_{t+1} - P_t\|_{\mathrm{op}} \|D_r\|_F \\ &\geq \Delta_0 - \|D_r\|_F \sum_{s=0}^{t} \|P_{s+1} - P_s\|_{\mathrm{op}}.\end{aligned}$$

This closes the induction for every finite t.

Each projector-motion term is nonnegative. Clause 2 therefore bounds every finite prefix by the full path budget:

$$\sum_{s=0}^{t-1} \|P_{s+1} - P_s\|_{\mathrm{op}} \leq \sum_{s=0}^{\infty} \|P_{s+1} - P_s\|_{\mathrm{op}} \leq L_P.$$

Clause 1 then yields

$$\begin{aligned}\mathrm{dist}_F(D_r, \mathcal{S}_t) &\geq \Delta_0 - L_P \|D_r\|_F \\ &\geq (\delta - L_P)\|D_r\|_F.\end{aligned}$$

No limit of P_t, $\mathcal{S}_t$, or the distance sequence is taken; the infinite series is used only as a finite-prefix budget. Finally,

$$\delta - L_P > \delta - \delta/4 = 3\delta/4 > 0.$$

Since $r \geq 1$, $\|D_r\|_F = \sqrt{r} > 0$. Thus a full coefficient span at any time, which would have zero distance, is ruled out as a conclusion of the two clauses rather than excluded as an assumption. Zero spans, stationary spans, and rank jumps remain covered by the charged recurrence. The projector defect may have the adversarial sign for a lower bound, but its total all-time charge is exactly at most $L_P \|D_r\|_F$; no cancellation or decay is invoked. The implication is deterministic on clauses 1 and 2. It does not assert that these clauses, or the full certificate, occur with positive probability. □

A.3 Transfer to an all-time ambient residual floor

Lemma A.5 (Distance to a subspace is one-Lipschitz)**.** *In the coefficient Frobenius space, let $\mathcal{S}$ be any linear subspace and let U, V be coefficient tensors. Then*

$$|\mathrm{dist}_F(U, \mathcal{S}) - \mathrm{dist}_F(V, \mathcal{S})| \leq \|U - V\|_F.$$

In particular,

$$\mathrm{dist}_F(U, \mathcal{S}) \geq \mathrm{dist}_F(V, \mathcal{S}) - \|U - V\|_F.$$

Proof. For every $S \in \mathcal{S}$, the triangle inequality gives

$$\|U - S\|_F \leq \|U - V\|_F + \|V - S\|_F.$$

Taking the infimum over $S \in \mathcal{S}$ yields

$$\mathrm{dist}_F(U, \mathcal{S}) \leq \|U - V\|_F + \mathrm{dist}_F(V, \mathcal{S}).$$

Interchanging U and V gives the reverse one-sided comparison, and the two inequalities prove the claim. The argument includes the zero subspace, the full coefficient space, and every rank-deficient intermediate subspace. The constant one is sharp when the perturbation points directly toward or away from the subspace. □

Proposition A.3 (Coefficient residual floor with explicit smoothing loss). *Under Assumption 3, Propositions A.1 and A.2, and the clause-4 bound*

$$\|E_\rho\|_F \leq \zeta \|D_r\|_F,$$

suppose also that $L_P < \delta/4$ and $\zeta < \delta/4$. Then every integer $t \geq 0$ obeys

$$\|Q(T - \widehat{T}_t)\|_F \geq (\delta - L_P - \zeta)\|D_r\|_F,$$

and

$$\delta - L_P - \zeta > \frac{\delta}{2} > 0.$$

Proof. Fix $t \geq 0$. Proposition A.1 gives

$$Q(T - \widehat{T}_t) = D_r + E_\rho - C_t, \qquad C_t \in \mathcal{S}_t.$$

Because C_t is an admissible point in $\mathcal{S}_t$,

$$\begin{aligned}\|Q(T - \widehat{T}_t)\|_F &= \|D_r + E_\rho - C_t\|_F \\ &\geq \operatorname{dist}_F(D_r + E_\rho, \mathcal{S}_t).\end{aligned}$$

Lemma A.5, with $U = D_r + E_\rho$ and $V = D_r$, gives

$$\operatorname{dist}_F(D_r + E_\rho, \mathcal{S}_t) \geq \operatorname{dist}_F(D_r, \mathcal{S}_t) - \|E_\rho\|_F.$$

Proposition A.2 and the displayed clause-4 bound now yield, without dropping either defect,

$$\begin{aligned}\|Q(T - \widehat{T}_t)\|_F &\geq (\delta - L_P)\|D_r\|_F - \zeta\|D_r\|_F \\ &= (\delta - L_P - \zeta)\|D_r\|_F.\end{aligned}$$

The strict restrictions give

$$\delta - L_P - \zeta > \delta - \frac{\delta}{4} - \frac{\delta}{4} = \frac{\delta}{2} > 0.$$

The time was arbitrary. Thus the floor holds at the raw entry and after every update. The smoothing residual is fixed and subtracted once in each pointwise comparison; it is not repeatedly accumulated along the horizon. An adversarially oriented E_ρ may consume the full ζ allowance, but no more. □

Proposition A.4 (Exact same-target ambient residual floor). *Under Assumption 3, Lemma A.2, Proposition A.3, and the remaining clause-4 condition*

$$\|T\|_F \leq C_T \|D_r\|_F, \qquad C_T > 0,$$

every integer $t \geq 0$ satisfies

$$\|T - \widehat{T}_t\|_F \geq \frac{\delta - L_P - \zeta}{\kappa^6 C_T}\|T\|_F.$$

The bound is deterministic, all-time, and has no hidden constant.

Proof. Fix $t \geq 0$ and abbreviate $R_t = T - \widehat{T}_t$ within this proof. Lemma A.2 gives

$$0 < \|Q\|_{\text{op}} \leq \kappa^6, \qquad \|QR_t\|_F \leq \|Q\|_{\text{op}}\|R_t\|_F.$$

Positivity permits division, while the upper bound on the operator norm gives the lower-bound direction

$$\|R_t\|_F \geq \frac{\|QR_t\|_F}{\|Q\|_{\mathrm{op}}} \geq \kappa^{-6}\|QR_t\|_F.$$

This inference needs no injectivity of Q and remains valid when the ambient residual has components in $\ker Q$. Proposition A.3 therefore yields

$$\|T - \widehat{T}_t\|_F \geq \kappa^{-6}(\delta - L_P - \zeta)\|D_r\|_F.$$

Since $C_T > 0$, the target-scale clause is equivalent to

$$\|D_r\|_F \geq C_T^{-1}\|T\|_F.$$

Substitution proves the stated bound. No division by $\|T\|_F$ occurs, so $T = 0$ is included and gives a zero right-hand side. Every scale factor is visible: L_P is the projector-path charge, ζ is the fixed smoothing loss, κ^6 is the coordinate distortion, and C_T is the target-scale conversion. There is no probability conversion and no time-dependent constant. □

Proposition A.5 (Exact/noiseless baseline residual floor)**.** *Under Assumption 3, consider the separate deterministic zero-smoothing baseline with* $n \geq r$ *and column-orthonormal matrices* $\bar{A}, \bar{B}, \bar{C}$*, all perturbation vectors equal to zero, and*

$$T = \sum_{j=1}^{r} \bar{a}_j \otimes \bar{b}_j \otimes \bar{c}_j.$$

Then

$$QT = D_r, \qquad E_\rho = 0, \qquad \|Q\|_{\mathrm{op}} = 1, \qquad \|T\|_F = \|D_r\|_F.$$

If clauses 1 and 2 of $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$ *hold with* $\delta > 0$*,* $L_P > 0$*, and* $L_P < \delta/4$*, then every* $t \geq 0$ *satisfies the stronger bound*

$$\|T - \widehat{T}_t\|_F \geq (\delta - L_P)\|T\|_F.$$

Proof. Column orthonormality gives $\Lambda_M = \bar{M}^\mathsf{T}$ and $\Lambda_M \bar{m}_j = e_j$ in each mode. Tensor-product linearity therefore gives

$$QT = \sum_{j=1}^{r} e_j \otimes e_j \otimes e_j = D_r,$$

so $E_\rho = QT - D_r = 0$. Lemma A.2 gives

$$\|Q\|_{\mathrm{op}} = \prod_{M \in \{A,B,C\}} \|\bar{M}^\mathsf{T}\|_{\mathrm{op}} = 1.$$

Moreover, the rank-one summands of T are orthonormal because, for every $j, \ell \in [r]$,

$$\left\langle \bar{a}_j \otimes \bar{b}_j \otimes \bar{c}_j, \bar{a}_\ell \otimes \bar{b}_\ell \otimes \bar{c}_\ell \right\rangle = \delta_{j\ell}^3.$$

Hence $\|T\|_F^2 = r = \|D_r\|_F^2$.

Proposition A.1 now gives the well-typed coefficient-space identity

$$Q(T - \widehat{T}_t) = D_r - C_t, \qquad C_t \in \mathcal{S}_t.$$

The operator relation $\|QR\|_F \le \|Q\|_{\mathrm{op}}\|R\|_F$ and $\|Q\|_{\mathrm{op}} = 1$ imply

$$\begin{aligned}
\|T - \widehat{T}_t\|_F &\ge \|Q(T - \widehat{T}_t)\|_F \\
&= \|D_r - C_t\|_F \\
&\ge \mathrm{dist}_F(D_r, \mathcal{S}_t) \\
&\ge (\delta - L_P)\|D_r\|_F \\
&= (\delta - L_P)\|T\|_F,
\end{aligned}$$

where Proposition A.2 gives the third inequality. Thus removing smoothing and coordinate distortion strengthens the same-target floor instead of replacing it by a conservative remainder, a stopped statement, or a vanishing-defect surrogate. □

A.4 Finite represented-tensor variation and objective convergence

Lemma A.6 (Finite variation gives ambient convergence)**.** *If clause 3 of $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$ holds, then $(\widehat{T}_t)_{t\ge 0}$ is Cauchy in ambient Frobenius norm. Hence there exists $\widehat{T}_\infty \in \mathbb{R}^{n\times n\times n}$ such that*

$$\widehat{T}_t \longrightarrow \widehat{T}_\infty \qquad \textit{in Frobenius norm.}$$

Proof. Define the increment norms and their tails by

$$v_j = \|\widehat{T}_{j+1} - \widehat{T}_j\|_F, \qquad V_N = \sum_{j=N}^{\infty} v_j.$$

Clause 3 states $V_0 < \infty$. Since the series is nonnegative, every tail is finite and $V_N \to 0$. For integers $s > t \ge 0$, exact telescoping gives

$$\widehat{T}_s - \widehat{T}_t = \sum_{j=t}^{s-1} (\widehat{T}_{j+1} - \widehat{T}_j),$$

and hence

$$\|\widehat{T}_s - \widehat{T}_t\|_F \le \sum_{j=t}^{s-1} v_j \le V_t. \tag{A.1}$$

Given $\varepsilon > 0$, choose N with $V_N < \varepsilon$. The tails are nonincreasing, so Equation (A.1) gives

$$\|\widehat{T}_s - \widehat{T}_t\|_F < \varepsilon \qquad \text{for all } s > t \ge N.$$

Thus the represented tensors are Cauchy. The ambient Frobenius space is isometric to finite-dimensional Euclidean space $\mathbb{R}^{n^3}$ and is therefore complete, which produces $\widehat{T}_\infty$ in the same tensor space. Letting $s \to \infty$ in Equation (A.1) also gives the useful tail estimate

$$\|\widehat{T}_\infty - \widehat{T}_t\|_F \le V_t \longrightarrow 0.$$

This derivation controls only represented tensors. It requires no factor bound or factor convergence, Gram conditioning, descent inequality, design-rank persistence, quotient geometry, or Kurdyka–Lojasiewicz property. The first increment v_0 is included. A stationary trajectory has $v_j = 0$ for every j and converges immediately; changing factor representatives or least-squares ranks does not affect the argument. □

Proposition A.6 (Finite objective limit)**.** *Under clause 3 of* $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$ *and Lemma A.6, the objective satisfies*

$$\mathcal{L}(X_t, Y_t, Z_t) \longrightarrow \|T - \widehat{T}_\infty\|_F^2 < \infty.$$

Proof. For fixed ambient tensors R, S, the Frobenius polarization identity and Cauchy–Schwarz imply

$$\big|\|T - R\|_F^2 - \|T - S\|_F^2\big| \le \|R - S\|_F \big(\|T - R\|_F + \|T - S\|_F\big).$$

Indeed, apply $\|A\|_F^2 - \|B\|_F^2 = \langle A - B, A + B\rangle$ to $A = T - R$ and $B = T - S$. Taking $R = \widehat{T}_t$ and $S = \widehat{T}_\infty$, and then using the triangle inequality, gives

$$\begin{aligned}
&\Big|\mathcal{L}(X_t, Y_t, Z_t) - \|T - \widehat{T}_\infty\|_F^2\Big| \\
&\quad \le \|\widehat{T}_t - \widehat{T}_\infty\|_F \big(\|T - \widehat{T}_t\|_F + \|T - \widehat{T}_\infty\|_F\big) \\
&\quad \le \|\widehat{T}_t - \widehat{T}_\infty\|_F \Big(2\|T - \widehat{T}_\infty\|_F + \|\widehat{T}_t - \widehat{T}_\infty\|_F\Big).
\end{aligned}$$

The first factor tends to zero by Lemma A.6; the second tends to the finite number $2\|T - \widehat{T}_\infty\|_F$. This proves convergence to the displayed finite value. The conclusion remains valid if the limiting objective is zero, if $T = 0$, if factors diverge while their represented tensor converges, or if the represented path is stationary. Mere finiteness of the variation series supplies no uniform numerical rate, and none is claimed. This is a deterministic implication on clause 3; it does not assign a probability to that clause or to the full certificate. ☐

A.5 Conditional theorem closure and the probability boundary

Lemma A.7 (Uniform lower bounds pass to a finite limit)**.** *Let* $(a_t)_{t\ge 0}$ *be a real sequence and let* $a, b \in \mathbb{R}$*. If*

$$a_t \longrightarrow a, \qquad a_t \ge b \quad \textit{for every } t \ge 0,$$

then $a \ge b$*.*

Proof. If instead $a < b$, set $\eta = (b - a)/2 > 0$. Convergence gives an integer N such that $|a_t - a| < \eta$ for every $t \ge N$. For such t,

$$a_t < a + \eta = \frac{a + b}{2} < b,$$

contradicting the pointwise lower bound. Thus $a \ge b$. The argument allows equality, constant sequences, and arbitrary real values of a and b; it uses an ordinary finite limit and makes no extended-real or convergence-mode upgrade. ☐

Proposition A.7 (Pathwise closure of the conditional loss floor)**.** *Under Assumptions 1, 2, 3, 4, and 5, suppose a realized trajectory belongs to* $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$*, where* $\delta, L_P, \zeta, C_T > 0$*,* $L_P < \delta/4$*, and* $\zeta < \delta/4$*. Define*

$$m = \delta - L_P - \zeta, \qquad \epsilon = \left(\frac{m}{\kappa^6 C_T}\right)^2.$$

Then $m > \delta/2 > 0$*,* $\epsilon > 0$*, and*

$$\lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) = \|T - \widehat{T}_\infty\|_F^2 < \infty,$$

with

$$\lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \ge \epsilon \|T\|_F^2.$$

Proof. The strict parameter restrictions give the complete margin calculation

$$m = \delta - L_P - \zeta > \delta - \frac{\delta}{4} - \frac{\delta}{4} = \frac{\delta}{2} > 0. \tag{A.2}$$

Since $\kappa \geq 1$ and $C_T > 0$, this also gives $\epsilon > 0$.

Membership in the four-clause certificate supplies the hypotheses of Propositions A.2 and A.4. Therefore, for every integer $t \geq 0$,

$$\|T - \widehat{T}_t\|_F \geq \frac{m}{\kappa^6 C_T} \|T\|_F. \tag{A.3}$$

Both sides of Equation (A.3) are nonnegative: the left side is a norm, while the right side is nonnegative by Equation (A.2), $\kappa \geq 1$, $C_T > 0$, and $\|T\|_F \geq 0$. Squaring only after this sign check and using the exact objective definition gives

$$\mathcal{L}(X_t, Y_t, Z_t) = \|T - \widehat{T}_t\|_F^2 \geq \left(\frac{m}{\kappa^6 C_T}\right)^2 \|T\|_F^2 = \epsilon \|T\|_F^2 \quad \text{for every } t \geq 0. \tag{A.4}$$

No term is absorbed or dropped in this equality.

The same certificate membership supplies clause 3. Lemma A.6 and Proposition A.6 give

$$\mathcal{L}(X_t, Y_t, Z_t) \longrightarrow \|T - \widehat{T}_\infty\|_F^2 < \infty.$$

Apply Lemma A.7 to this scalar sequence and the all-time lower bound in Equation (A.4). The limiting lower bound follows.

The primitive dimension, rank-window, smoothing, and initialization assumptions preserve the exact theorem domain and joint law; they are not used to prove certificate membership. If $T = 0$, the proof never divides by $\|T\|_F$ and the lower bound has zero right-hand side. Stationary paths are covered by the finite-variation argument. No Gram conditioning, factor boundedness or convergence, descent property, fixed span, basin membership, design-rank persistence, quotient geometry, or unstated feature of the ALS map enters the closure. □

Proposition A.8 (Exact conditional event inclusion)**.** *Under Assumptions 1, 2, 3, 4, and 5, let the positive certificate parameters satisfy $L_P < \delta/4$ and $\zeta < \delta/4$, and define*

$$\epsilon = \left(\frac{\delta - L_P - \zeta}{\kappa^6 C_T}\right)^2.$$

Then, under the joint smoothing-and-initialization law,

$$\mathsf{C}_2(\delta, L_P, \zeta, C_T) \subseteq \left\{ \begin{array}{l} \lim\limits_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \text{ exists and is finite,} \\ \lim\limits_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \geq \epsilon \|T\|_F^2 \end{array} \right\}.$$

The inclusion asserts neither nonemptiness nor positive probability of the antecedent event.

Proof. Take an arbitrary outcome of the joint experiment that belongs to $\mathsf{C}_2(\delta, L_P, \zeta, C_T)$. Proposition A.7 applies to the realized target and trajectory and proves both properties defining the event on the right. Thus every element of the left-hand set is an element of the right-hand set.

This is an outcome-by-outcome implication inside the primitive probability space. It does not invoke a conditional probability, divide by $\mathbb{P}[\mathsf{C}_2]$, or infer a lower bound on that probability. It therefore remains valid when the certificate is empty or has probability zero. The probability mode is exactly set inclusion, not high probability, expectation, or an unconditional ALS success statement. □

Proposition A.9 (Exact/noiseless limiting-loss baseline). *Consider the separate deterministic zero-smoothing baseline with $n \geq r$, column-orthonormal matrices $\bar{A}, \bar{B}, \bar{C}$, all perturbation vectors equal to zero, and*

$$T = \sum_{j=1}^{r} \bar{a}_j \otimes \bar{b}_j \otimes \bar{c}_j.$$

Thus $QT = D_r$, $E_\rho = 0$, $\|Q\|_{\mathrm{op}} = 1$, and $\|T\|_F = \|D_r\|_F$. Suppose $\delta > 0$, $L_P < \delta/4$, and the trajectory satisfies the normalized entry-deficit, finite projector-path, and finite represented-tensor-variation clauses. Then

$$\lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) = \|T - \widehat{T}_\infty\|_F^2 < \infty$$

and

$$\lim_{t\to\infty} \mathcal{L}(X_t, Y_t, Z_t) \geq (\delta - L_P)^2 \|T\|_F^2.$$

Proof. Proposition A.5 gives, for every $t \geq 0$,

$$\|T - \widehat{T}_t\|_F \geq (\delta - L_P)\|T\|_F.$$

The restriction $L_P < \delta/4$ gives $\delta - L_P > 3\delta/4 > 0$, so both sides are nonnegative and may be squared:

$$\mathcal{L}(X_t, Y_t, Z_t) \geq (\delta - L_P)^2 \|T\|_F^2 \quad \text{for every } t \geq 0.$$

The finite-variation clause and Proposition A.6 give the displayed finite objective limit. Lemma A.7 passes the pointwise baseline floor to that same limit. Hence the deterministic zero-smoothing tall partial-isometry baseline preserves the stronger constant relative floor; it is not replaced by an absolute remainder, a stopped statement, or a claim that only defect terms vanish. This conclusion is algebraic and is not a probability statement under the positive-Gaussian-smoothing law. □

A.6 Proof of the Main Theorem

Proof of Theorem 3.1. Fix κ, q and any fixed constants with the dependence and strict margins stated in Theorem 3.1. For arbitrary dimensions, base triple, target, and ALS trajectory in the theorem scope, Proposition A.1 proves the exact coordinate identity and the bound $\|Q\|_{\mathrm{op}} \leq \kappa^6$. Proposition A.2 converts the first two certificate clauses into an all-time coefficient deficit. Proposition A.4 uses the fourth clause to transfer that deficit to the exact ambient residual with factor

$$\frac{\delta - L_P - \zeta}{\kappa^6 C_T}.$$

Independently, Proposition A.6 uses the third clause to produce a finite limit of the same objective sequence. Proposition A.8 combines those conclusions and gives precisely the event inclusion and the stated value of ϵ.

The displayed constants have no hidden dependence. The residual inequality is valid for every time, while the conclusion concerns the asymptotic objective; the scalar passage between these modes is Lemma A.7. The argument is deterministic on the certificate and performs no probability conversion.

Because the implication holds for every positive constant choice satisfying the two strict margins, the existential formulation is nonempty at the scalar level. For example, one may take

$$r_0 = 1, \qquad C_{\mathrm{dim}} = 1, \qquad \delta = 1, \qquad L_P = \zeta = \frac{1}{8}, \qquad C_T = 1,$$

and define ϵ by the theorem formula. These are constant functions of (κ, q) and hence have the permitted dependence. This choice only discharges the existential quantifiers over scalar constants; it does not assert that the resulting certificate event is nonempty or has positive probability.

Finally, move outside the positive-Gaussian-smoothing law to the separate deterministic baseline stated in Theorem 3.1: the tall base matrices have orthonormal columns, all perturbation vectors vanish, and the target is their deterministic rank-r CP tensor. The resulting relations $QT = D_r$, $E_\rho = 0$, $\|Q\|_{\mathrm{op}} = 1$, and $\|T\|_F = \|D_r\|_F$ are verified in Proposition A.5, which gives the typed all-time residual floor from the same-target, projector-path, and operator relations. Proposition A.9 combines that floor with finite variation and proves the stated algebraic limiting-loss conclusion with factor $(\delta - L_P)^2$. No probability claim is made for this zero-smoothing comparison. This completes every part of Theorem 3.1. $\square$